

Managing Networks for LLM Training

Research Paper Summary Report

This report summarizes recent research papers related to network management, congestion control, fault tolerance, routing optimization, and infrastructure resilience for Large Language Model (LLM) training. The focus is on distributed GPU clusters, RDMA networking, traffic optimization, and simulation frameworks used in large-scale AI training systems.

1. LCMP: Distributed Long-Haul Cost-Aware Multi-Path Routing for Inter-Datcenter RDMA Networks (2026)

- Introduces a distributed routing framework for RDMA-based LLM training across datacenters.
- Uses path-quality scoring and congestion-aware routing to reduce latency.
- Improves median and tail Flow Completion Time (FCT) by up to 76% and 64%.
- Demonstrates scalability across 8-datcenter testbeds and NS-3 simulations.

Why it matters: Important for reducing communication bottlenecks and optimizing large-scale distributed LLM training.

2. RAPID-LLM: Resilience-Aware Performance Analysis of Infrastructure for Distributed LLM Training and Inference (2025)

- Combines DeepFlow trace generation with Astra-Sim backend for infrastructure modeling.
- Supports congestion-aware routing and degraded/faulty links.
- Predicts training latency within 10.4% accuracy compared to real deployments.
- Enables simulation of hybrid parallelism and network failures.

Why it matters: Useful for evaluating network reliability and scalability before deployment.

3. SHIFT: An RDMA Failure-Resilient Layer for Distributed Training (2025)

- Introduces RDMA-layer fault tolerance for distributed AI training.
- Allows seamless rerouting of traffic across NICs without stopping training.
- Works transparently with frameworks like PyTorch and NCCL.
- Minimizes computation loss caused by network failures.

Why it matters: Critical for maintaining training continuity in large GPU clusters.

4. Reunion: Receiver-driven Network Load Balancing Mechanism in AI Training Clusters (2025)

- Focuses on elephant-flow congestion in RoCEv2 AI clusters.
- Uses Count-Min-Sketch and real-time congestion data for dynamic rerouting.

- Reduces tail latency by up to 62.1% compared to ECMP and ConWeave.
- Addresses packet reordering and flowlet timeout challenges.

Why it matters: Helps optimize collective communication operations like AllReduce.

5. Congestion Patterns in a Large-scale RDMA Datacenter (2025)

- Analyzes congestion behavior in AI-focused RDMA datacenters.
- Finds congestion shifts from edge-hosts to network cores under PFC.
- Highlights burstiness and synchronization issues in AI traffic.
- Provides insights into telemetry and congestion observability.

Why it matters: Provides operational insights for monitoring and congestion management.

6. Holmes: Towards Distributed Training Across Clusters with Heterogeneous NIC Environment (2023)

- Enables LLM training across clusters with mixed NIC technologies.
- Introduces intelligent task scheduling based on NIC capability.
- Improves training efficiency compared to pure Ethernet environments.
- Supports scalable pipeline parallelism across multiple clusters.

Why it matters: Useful when training infrastructure contains heterogeneous networking hardware.

7. Vela: A Virtualized LLM Training System with GPU Direct and RoCE (2025)

- Cloud-native architecture for large-scale LLM training using RoCE.
- Supports GPU Direct RDMA and virtualization with KVM.
- Achieved ~80% ideal throughput on 1500 GPUs.
- Highlights practical deployment lessons for AI infrastructure.

Why it matters: Shows how virtualization can still provide high-performance networking for LLM training.

8. Nonuniform-Tensor-Parallelism: Mitigating GPU Failure Impact for Scaled-up LLM Training (2025)

- Addresses GPU failures in large tensor-parallel training jobs.
- Introduces adaptive tensor-parallel execution with reduced TP degree.
- Maintains near-zero throughput loss during partial GPU failures.
- Combines power-boosting strategies with fault-tolerant training.

Why it matters: Important for improving resilience in extremely large GPU deployments.

Overall Findings:

The current direction of LLM networking research strongly emphasizes congestion-aware routing, RDMA optimization,

fault resilience, simulation-driven evaluation, and scalable parallelism. The most impactful techniques involve adaptive traffic engineering, intelligent load balancing, and predictive failure handling to reduce GPU idle time and network bottlenecks. Tools such as Astra-Sim, Chakra traces, and resilience-aware simulators are increasingly important for testing future AI infrastructure before real-world deployment.

References

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